

# Planning Document - Control Board

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## Goal

Control Board for 2027 Revision of RJ RoboCup Robots. Radio has been offloaded to [2027 Radio Module Design Doc](#) that will slot into the control board.

Associated STM32H723ZGT6 devkit: [NUCLEO-H723ZG | Product - STMicroelectronics](#)

## Requirements

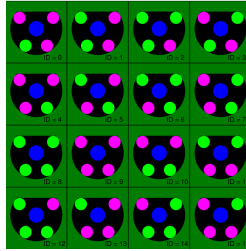
### 1. Power Input/Output

- ☐ **3.3 V Rail Input @ 1 A → Supposedly changing this to 2 A**
  - ☐ **Current estimated total: 1 A**
    - ☐ **150 mA** from STM32H723ZGT6
    - ☐ **650 mA** from Radio Module (ESP32-C5 sucks up A LOT)
    - ☐ **~200 mA** from SD Card
- ☐ **5 V Rail Input @ 1 A**
  - ☐ **Current estimated total: 330 mA**
    - ☐ **60 mA \* 5 = ~300 mA** from 5 Dotstar LEDs
    - ☐ **29 mA** from OLED Display
    - ☐ Rest goes to I2C Expansion ports?

### 2. Major Features

- ☐ **STM32H723ZGT6** main MCU chip at up to 550 Mhz (defaulting to 250 Mhz for now)
- ☐ Must be easily programmable
  - ☐ **USB-C is preferred** (onboard ST-Link or [STM32 USB DFU Bootloader: Flash Firmware Over USB Without ST-Link](#))
  - ☐ Allowing easy ST-Link plugin is acceptable ([5 pin SWD for reference](#))
- ☐ 6 position DIP switch (for Robot ID or other config/debug). Can be routed to any free GPIO.
- ☐ SD Card compatibility via 4-bit SDMMC
- ☐ Screwable Display
  - ☐ [326 Adafruit Industries LLC | LCD, OLED, Graphic | DigiKey](#)
    - ☐ This is supported by the u8g2 library, so easier porting from last rev.
  - ☐ Requires 4x M2-sized PCB screwholes
  - ☐ Requires I2C (400 Khz Fast Mode, or 100 Khz Standard Mode)
    - ☐ Recommended to use ~3 kΩ pull-up resistors on both SDA and SCL lines (value depends on trace length)
    - ☐ SPI available but unnecessary for use-case
- ☐ Motorboards over shared SPI line (Prescaled down to 15.625 Mb/s)
- ☐ Kicker over I2C (400 Khz Fast Mode, or 100 Khz Standard Mode)
- ☐ Open expansion I2C ports, 5V

- ☐ Connection to ESP-C5 Radio Daughterboard via Standard SPI + ESP-hosted int/rst pins
- ☐ 2 expansion JST slots
  - ☐ 5V Power and I2C (400 Khz Fast Mode, or 100 Khz Standard Mode)
- ☐ [BB-2020BGR-TRB](#) 5 Dotstar Addressable LEDs



- ☐ Should be placed to mimic the 5 dot pattern of robot shells
- ☐ Uses standard SPI (Prescaled down to 7.815 Mbits/s to meet 12 Mhz absolute limit of DotStar LEDs)
- ☐ 2 simple red LED lights indicating 5V and 3V rail are active
- ☐ [PTS820](#) 3 programmable GPIO buttons
  - ☐ Please try to add some hardware debounce!
- ☐ [BMI088](#) IMU via I2C

### 3. Constraints

- ☐ I2C lines cannot be shared between peripherals
  - ☐ For example, the OLED's I2C should be on a different line than the Kicker I2C
  - ☐ Exception: JST expansion slots can share the same I2C hardware
- ☐ Make sure ports are using secure connectors (try to find cheaper Molex alternatives)

### 4. Connectors ([pinouts here](#))

- ☐ See [PDF ControlBoard\\_Testing\\_V0.2.pdf](#) for pinouts out of STM32. This pin config will also be viewable within the new firmware codebase.

## Critical Parts

Parts that contribute significantly to meeting requirements.

### MCU: STM32

- Purchase Link: XXX
- Datasheet: XXX
- Design Guide: XXX
- Design Considerations
  - Must XXX

XXX

- XXX